

Validation: V_k and dV_k for $k_p dt$ in (0, 2), unit initial error

$y^2 - 1) V < 0$ proved symbolically (CONTROL-LYAPUNOV-001);

shown at machine precision for illustration. $k_p dt = 1$ converges in one step; 1.9 oscillates but contracts.

Symbolically verified under stated assumptions; trajectories illustrative

| Wolfram 15.0.0 | WorkingPrecision 50 | cross-language tol 1e-9 | 5 gain values, 30 steps

